MILITARY SPECIFICATION

ALUMINUM HYDROXIDE GEL, DRIED, MAGNESIUM TRISILICATE, AND SODIUM BICARBONATE TABLETS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Aluminum Hydroxide Gel, Dried, Magnesium Trisilicate and Sodium Bicarbonate Tablets (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (See 6.3).

STANDARDS

Federal

FED-STD-140	Tablets (for Medicinal Purposes).

FSC 6505

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Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Aluminum Hydroxide Gel, Dried, Magnesium Trisilicate and Sodium Bicarbonate Tablets, with Alginic Acid and shall be in accordance with all applicable requirements of FED-STD-140 and as specified herein:

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S2. Classification. Shall be type I, Class 1.

S5.3 The following additional requirements and tests are added to this paragraph:

Shall be pleasantly flavored, chewable tablets containing in each tablet the following:

Aluminum Hydroxide Gel, Dried	80 mg
Magnesium Trisilicate	20 mg
Sodium Bicarbonate	70 mg
Alginic Acid	200 mg

The tablets shall comply with the following:

Shall contain not less than 90.0 percent and not more than 110.0 percent of the required amount of Dried Aluminum Hydroxide Gel when determined as specified in 4.3.1.

Shall contain not less than 90.0 percent and not more than 110.0 percent of the required amount of Magnesium Trisilicate when determined as specified in 4.3.2.

Shall contain not less than 90.0 percent and not more than 110.0 percent of the required amount of Sodium Bicarbonate when determined as specified in 4.3.3.

Shall contain not less than 90.0 percent and not more than 110.0 percent of the required amount of Alginic Acid when determined as specified in 4.3.4.

The Dried Aluminum Hydroxide Gel, Magnesium Trisilicate and Sodium Bicarbonate used in the manufacture of the tablets shall be in accordance with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

The Alginic Acid used in the manufacture of the tablets shall comply with the latest edition of the Food Chemicals Codex.

All other ingredients used in the manufacture of the tablets shall comply with S5.1.

S6.4.3 Flavored tablets. The tablets shall be flavored, and shall be palatable and pleasant to the taste with no unpleasant after-taste. The tablets shall comply with the palatability test as specified in 4.3.5.

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S6.4.6 Hardness. The tablets shall have a hardness of 7.0 Kg to 10.0 Kg when tested with the Pfizer Hardness Tester or as an alternate the tablets shall have a hardness of 8.0 kp to 12.0 kp when tested with the Heberlein Hardness Tester. Not less than twenty (20) tablets shall be tested.

S6.4.9 Disintegration and solubility. Tablets shall meet the requirements of the U.S.P. tablet disintegration test for uncoated tablts, using simulated gastric fluid, in not more than 30 minutes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. Sampling shall be conducted in accordance with FED-STD-140.

4.3 Tests.

4.3.1 Dried Aluminum Hydroxide Gel.

Reagents: 6 N Hydrochloric Acid
Aluminum Standard 1000 ppm Aluminum (Available from Fisher Scientific)

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Apparatus: Atomic Absorption Spectrophotometer, Perkin Elmer 303 (or equivalent instrument yielding comparable results) using the following operating conditions:

Wavelength	309 UV
Slit	4
Oxidant	Nitrous Oxide
Fuel	Acetylene

Procedure:

Weigh and finely powder twenty tablets. Pass the powdered tablets through a 40 mesh screen. Accurately weigh approximately 1.0 gram of the powdered tablets and transfer to a 100 ml volumetric flask. Add 10 ml of 6 N HCl. Heat on a steam bath for fifteen minutes, cool, and q.s. to volume with purified water. Filter through #42 filter paper, discarding the first 20 ml of filtrate. Transfer 15 ml of the filtrate into a 100 ml volumetric flask and q.s. to volume with purified water. Determine the absorption of the sample against purified water using the atomic absorption apparatus. The unit shall be calibrated for aluminum by transferring 4 and 5 ml of the aluminum standard solution to separate 100 ml volumetric flasks and diluting to volume with purified water. Determine the absorbance of the two solutions (40 and 50 PPM) against purified water.

Calculate the quantity of aluminum and the quantity of Dried Aluminum Hydroxide Gel in the sample.

4.3.2 Magnesium Trisilicate.

Reagents: 6 N Hydrochloric Acid
Magnesium Standard — 1000 ppm Magnesium (Available from Fisher Scientific)

Apparatus: Atomic Absorption Spectrophotometer, Perkin Elmer 303 (or equivalent instrument yielding comparable results) using the following operating conditions:

Wavelength	285 UV
Slit	4
Oxidant	Nitrous Oxide
Fuel	Acetylene

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Procedure:

Weigh and finely powder twenty tablets. Pass the powdered tablets through a 40 mesh screen. Accurately weigh approximately 1.0 gram of the powdered tablets and transfer to a 100 ml volumetric flask. Add 10 ml of 6 N HCl. Heat on a steam bath for fifteen minutes, cool, and q.s. to volume with purified water. Filter through #42 filter paper, discarding the first 20 ml of filtrate. Transfer 10 ml of the filtrate into a 100 ml volumetric flask and q.s. to volume with purified water. Determine the absorption of the sample against purified water using the atomic absorption apparatus. The unit shall be calibrated for Magnesium by transferring 10 ml of the Magnesium standard solution to a 100 ml volumetric flask and diluting to volume with purified water. Transfer 17 and 18 ml aliquots of this solution to separate 100 ml volumetric flasks, add 1.0 ml of 6 N HCl and q.s. to volume with purified water (1.7 and 1.8 ppm). Determine the absorbance of the two solutions against purified water.

Calculate the quantity of Magnesium and the quantity of Magnesium Trisilicate in the sample.

4.3.3 Sodium Bicarbonate.

Reagents: 6 N Hydrochloric Acid
Sodium Standard — 1000 ppm Sodium (Available from Fisher Scientific)

Apparatus: Atomic Absorption Spectrophotometer, Perkin Elmer 303 (or equivalent instrument yielding comparable results) using the following operating conditions:

Wavelength	295-VIS
Slit	4
Oxidant	Air
Fuel	Acetylene

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Procedure:

Weigh and finely powder twenty tablets. Pass the powdered tablets through a 40 mesh screen. Accurately weigh approximately 1.0 gram of the powdered tablets and transfer to a 100 ml volumetric flask. Add 10 ml of 6 N HCl. Heat on a steam bath for forty-five minutes, cool, and q.s. to volume with purified water. Filter through #42 filter paper, discarding the first 20 ml of filtrate. Transfer 5 ml of the filtrate into a 500 ml volumetric flask and q.s. to volume with purified water. Determine the absorption of the sample against purified water using the atomic absorption apparatus. The unit shall be calibrated for Sodium by transferring 10 ml of the Sodium standard solution to a 100 ml volumetric flask and diluting to volume with purified water. Transfer 10 ml of this solution to another 100 ml volumetric flask and q.s. with purified water. Transfer 11 and 12 ml aliquots of this solution to separate 100 ml volumetric flasks, add 0.1 ml of 6 N HCl and q.s. to volume with purified water (1.1 and 1.2 ppm). Determine the absorbance of the two solutions against purified water.

Calculate the quantity of Sodium and the quantity of Sodium Bicarbonate in the sample.

Note: Throughout the assay, take adequate measures to avoid introducing any sodium contaminant into the standard or sample by the means of the glassware or handling the tablets.

4.3.4 Alginic Acid.

Procedure:

Weigh and grind twenty tablets. Accurately weigh a portion of ground tablets equivalent to approximately 100 mg of Alginic acid and transfer to a 50 ml centrifuge tube. Add 20 ml of purified water and shake for one minute. Accurately weigh about 100 mg of alginic acid standard into a 50 ml centrifuge tube. Add 20 ml purified water and shake for one minute. Treat the standard and sample as follows:

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Add 5 ml of 10% NH_3 solution. Mix for 1 minute on a Vortex or equivalent mixer. Filter the sample through a Gooch crucible with a thin asbestos mat (use long fiber asbestos) into a 125 ml flask. Transfer the filtrate into a 50 ml volumetric flask with the aid of 2% NH_3 solution. Q.s. to volume with 2% NH_3 solution and mix.

Transfer a 5 ml aliquot into a 35 ml centrifuge tube. Add 3 ml of purified water and 2 ml of 12 N Sulfuric Acid. Shake for 5 minutes. Centrifuge for 15 minutes. Aspirate the supernatent and discard.

Add 10 ml of 0.1 N Sulfuric Acid and mix for one minute on the Vortex mixer. Centrifuge for 15 minutes. Aspirate the supernatent and discard. Repeat this step, adding 10 ml of 0.1 N Sulfuric Acid etc.

Add 5 ml of 2% NH_3 solution to dissolve the alginic acid and transfer to a 10 ml volumetric flask. Wash the tube well with 2% NH_3 and q.s. to volume.

Determine the absorbance of the sample and standard solutions at 215 nm using a suitable spectrophotometer, 1 cm cells, and 2 percent ammonia as a blank.

Calculate the amount of Alginic Acid per tablet.

4.3.5 Palatability test. A taste panel consisting of 10 members will be used to determine acceptability of samples. Samples will be prepared for testing (samples will be tested undiluted), coded, and served to panel members under controlled serving conditions, e.g., all samples will be of the same amount, and served at the same temperature; each panel member will receive an equal number of samples; the order of serving will be varied among panel members; an interval of at least five (5) minutes will elapse between successive samples and panel members will rinse their mouths with water (room temperature) after each sample; panel members will test without interference either from each other or from outsiders. The product offered shall be rated equal to or better than the Palatability Standard,* when determined by the taste panel, using the following 9-point hedonic rating scale. The average rating of the samples shall be equal to or greater than the average rating of the standard, similarly prepared and tested.

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1	2	3	4	5	6	7	8	9
Dislike extremely	Dislike very much	Dislike moderately	Dislike slightly	Neither like nor dislike	Like slightly	Like moderately	Like very much	Like extremely

* The Palatability Standard is available, upon separate request to the contracting officer, Defense Personnel Support Center, Directorate of Medical Materiel.

4.3.5.1 Samples. Not later than the time specified for opening the bids or receipt of proposals, the offeror shall submit to the contracting officer 18 units of product which are representative of the product which the offeror proposes to furnish. A portion of the samples will be subjected to panel testing for a determination of palatability (see Palatability test). A portion will be used by the cognizant Government inspection and quality assurance activities for determining compliance of supplies furnished hereunder with the palatability requirement. Approval as to palatability of any sample submitted by the offeror will not constitute approval of the sample as to any other requirement of this specification. The requirement for submission of samples for use in determining compliance with the palatability requirement may be waived, provided the offeror states, in his bid or proposal, that the product he proposes to furnish is the same product he has offered to the purchasing activity on a previous procurement and the contracting officer determines that such product was previously procured and/or tested by the purchasing activity and found to comply with the specification requirement.

4.4 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table I. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units. For labeling, the acceptance number shall be zero.

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Table I. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing one hundred (100) tablets as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. Immediate container shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue shall be packaged as specified in 5.2.8 of PPP-C-186.

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5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Seventy-two (72) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Seventy-two (72) units of issue shall be packed as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the date of manufacture prefixed by "MFD"

5.4.1.2 Unit package. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.
(b) the National Stock Number.

Notes: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied the National Stock Number shall appear on the immediate container.

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5.4.2 Intermediate package. Each intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.3 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock No.	Item Identification
6505-00-148-4631	ALUMINUM HYDROXIDE GEL, DRIED, MAGNESIUM TRISILICATE AND SODIUM BICARBONATE TABLETS, 100s

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superceded by a revision.

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1385
cs

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-A-37230 ALUMINUM HYDROXIDE GEL, DRIED, MAGNESIUM

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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